Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Biosensor-Integrated 3D-Printed Microfluidic Tumor-on-a-Chip Systems as an Alternative to Preclinical Animal Testing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Clinical Chemistry & Biological Analysis Instruments	Medical Bio	Bio-Artificial Organs
Concept of Proposed Technology	The proposed technology is a biosensor-integrated Tumor-on-a-Chip (ToC) platform designed to replicate key biological, chemical, and physical characteristics of the human tumor microenvironment under controlled microfluidic conditions. The platform combines additively manufactured microfluidic chip architectures, living human tumor cells, tumor-biomimetic hydrogel matrices, and embedded metabolic biosensors into a single, integrated system. Microfluidic chips are fabricated using high-resolution Digital Light Processing (DLP) 3D printing, enabling precise control over channel geometry, perfusion paths, and cell culture compartments. The chips are produced from biocompatible and optically transparent photopolymer materials, allowing real-time optical monitoring and compatibility with standard bioimaging techniques.			

	Human tumor cells are embedded in tailored hydrogel matrices composed of biopolymers such as collagen, gelatin, and hyaluronic acid. These hydrogels reproduce essential biochemical and mechanical properties of the extracellular matrix, supporting cell viability, spatial organization, and biologically relevant tumor behavior during long-term culture. To enable continuous, non-invasive monitoring, nanoparticle-based oxygen and pH biosensors are directly integrated into the hydrogel and microfluidic environment. These biosensors are sterilizable, optically readable, and provide real-time, spatially resolved information on tumor metabolism, enabling dynamic observation of cellular responses to drug exposure and environmental changes. Overall, the technology provides a human-relevant, scalable, and digitally compatible alternative to conventional animal-based preclinical models, with direct applicability in oncology research, drug development, toxicity screening, and personalized medicine.
Rationale for Joint R&D Project	The development of an integrated Tumor-on-a-Chip platform requires the convergence of advanced microfluidic engineering, functional materials, biosensor technology, and biological validation, which cannot be efficiently achieved by a single organization. Joint R&D collaboration is essential for the following reasons: • Microfluidic system design and manufacturability require industrial expertise in precision fabrication, additive manufacturing, and scalable production. • Sterilizable, nanoparticle-based biosensors must be co-developed and optimized for compatibility with living biohybrid systems and microfluidic environments. • Tumor-biomimetic hydrogels must be jointly tailored to ensure compatibility with both biological requirements and chip material properties. • Early integration of industrial partners is critical to ensure commercial feasibility, quality control, and regulatory readiness. Collaboration with Korean SMEs enables: • Adaptation of the technology to industrial manufacturing and quality standards • Development of market-ready and application-oriented prototypes • Acceleration of technology transfer, certification preparation, and commercialization Through international joint R&D, the project aims to bridge laboratory innovation and industrial implementation, ensuring that the developed technology is scientifically robust, industrially viable, and aligned with global trends toward animal-free testing.
Key Words	1. Tumor-on-a-Chip 2. Microfluidic biomedical systems 3. DLP-based 3D printing 4. Biosensor integration (O_2, pH) 5. Animal-free preclinical testing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5-6)
R&D Objectives	The primary objective is to develop a biosensor-enabled Tumor-on-a-Chip platform that can serve as a reliable and human-relevant alternative to animal experiments in preclinical oncology research. Specific objectives are to: • Design and simulate microfluidic chip architectures that reproduce physiologically relevant tumor microenvironments • Develop biocompatible, optically transparent, and sustainable photopolymer materials suitable for high-resolution DLP 3D printing • Develop sterilizable nanoparticle-based oxygen and pH biosensors compatible with long-term biological integration • Develop tumor-biomimetic hydrogel matrices for simultaneous cell culture and sensor embedding • Fabricate and integrate functional Tumor-on-a-Chip prototypes • Demonstrate reproducibility, long-term stability, and biological relevance under microfluidic conditions			
R&D Target Specifications	• Microfluidic systems with fine channels (working volume: ≤ 100 μL) • Optical transparency: compatible with fluorescence microscopy and CLSM • Biosensor functionality: real-time monitoring of oxygen concentration and pH • Sterilization compatibility: ethanol, UV, and autoclave procedures • Biological performance: stable long-term tumor cell culture under continuous perfusion • Manufacturing approach: reproducible DLP-based fabrication of microfluidic chips			
R&D Plan	The R&D activities are structured into the following phases: 1. Microfluidic design and simulation: CAD-based chip design and multiphysics simulations to optimize flow profiles, nutrient transport, and shear stress. 2. Material and hydrogel development: Development and biocompatibility testing of biobased photopolymers and tumor-biomimetic hydrogel systems. 3. Biosensor development: Synthesis, functionalization, optical characterization, and sterilization testing of nanoparticle-based oxygen and pH sensors. 4. Chip fabrication and system integration: DLP-based fabrication of microfluidic chips and integration of hydrogels, tumor cells, and biosensors. 5. Biological validation and benchmarking: Long-term culture studies, sensor calibration, and comparison with conventional in-vitro methods and available reference data.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development and scaling of biocompatible DLP-printable materials • Development and optimization of tumor-biomimetic hydrogel matrices • Development and scaling of sterilizable nanoparticle-based oxygen and pH biosensors • Biocompatibility and material performance evaluation • Fabrication and optimization of chip prototypes • Data management and FAIR-compliant digital integration • Support for regulatory strategy and certification preparation
Expected R&R of Korean SME Partner	• Design and engineering of microfluidic chip architectures • Development and optimization of sterilizable nanoparticle-based oxygen and pH biosensors • Fabrication and optimization of Tumor-on-a-Chip prototypes • Integration testing, performance validation, and quality assurance • Biological testing, long-term cell culture, and functional validation • reparation for industrial scale-up and commercialization • Regulatory strategy and certification preparation for Korea/Asia

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated, biosensor-integrated Tumor-on-a-Chip prototype capable of real-time metabolic monitoring of human tumor models under physiologically relevant microfluidic conditions. • Expected outcomes and impacts include: • Reduction and partial replacement of animal experiments in preclinical oncology research • Improved predictability of drug efficacy and toxicity compared to animal models • Shortened development timelines and reduced costs in preclinical testing • Creation of new commercialization opportunities for SMEs in biotechnology, pharmaceuticals, and medical devices • Contribution to ethical, sustainable, and human-relevant biomedical research methodologies The technology is aligned with international regulatory trends and industrial demand for animal-free testing platforms and supports the global transition toward advanced in-vitro evaluation systems.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization